

AI-Based Weather Prediction System

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Abstract

Weather prediction is an important application of Artificial Intelligence and Machine Learning because accurate weather information supports agriculture, transportation, disaster management, water resource planning, and daily decision-making.

This research presents the development of an AI-based Weather Prediction System for predicting temperature using relevant meteorological parameters. The system follows a complete machine learning workflow including data collection, data preprocessing, exploratory data analysis, feature selection, model training, evaluation, and application development.

The proposed system is implemented using Python and machine learning libraries. Humidity, wind speed, and atmospheric pressure are used as input features for temperature prediction. A Random Forest Regressor is trained using the selected features. The trained model is integrated into a user-friendly application where users can enter weather parameters and obtain a predicted temperature.

The model is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Keywords: Weather Prediction, Machine Learning, Artificial Intelligence, Temperature Prediction, Meteorological Data, Random Forest, Python

1 Introduction

Weather conditions influence many aspects of human life and economic activity. Accurate weather prediction is particularly important for agriculture, transportation, disaster preparedness, energy management, and water resource planning.

Traditional weather forecasting methods depend on physical models, historical observations, and numerical weather prediction techniques. With the availability of large

meteorological datasets, Artificial Intelligence (AI) and Machine Learning (ML) provide additional approaches for identifying patterns in weather data.

Machine learning algorithms can learn relationships between meteorological variables from historical observations and use these relationships to make predictions for new observations. This research focuses on the development of an AI-based Weather Prediction System that predicts temperature from selected weather parameters.

The project follows a data-driven approach consisting of data preparation, exploratory analysis, model development, model evaluation, and application development. The final system provides a simple interface through which users can enter weather-related values and obtain a temperature prediction.

2 Problem Statement

Weather variables are influenced by several interacting atmospheric factors, making accurate prediction a challenging task. Conventional forecasting approaches can require complex numerical models, extensive computational resources, and specialized meteorological infrastructure.

The objective of this work is to develop a machine learning based system that can learn relationships between available meteorological parameters and temperature. The system processes weather data, trains a predictive model, evaluates its performance, and provides predictions through an easy-to-use application.

3 Objectives

The major objectives of the proposed system are:

- To study the application of Artificial Intelligence and Machine Learning in weather prediction.
- To collect and prepare meteorological data for analysis.
- To perform data preprocessing and exploratory data analysis.
- To identify relevant weather parameters for temperature prediction.
- To train a machine learning regression model.
- To evaluate the prediction model using suitable performance metrics.
- To develop a user-friendly weather prediction application.
- To document the complete methodology and experimental results.

4 Literature Review

Machine learning has increasingly been applied to weather and climate prediction because of its ability to identify relationships in large datasets. Previous research has explored regression algorithms, decision-tree-based methods, neural networks, recurrent neural networks, and time-series forecasting techniques for predicting meteorological variables.

Regression-based machine learning approaches are commonly used when the target variable is continuous, such as temperature, rainfall amount, or atmospheric pressure.

Ensemble methods such as Random Forest combine multiple decision trees and can model nonlinear relationships between input variables and the target variable. This makes Random Forest a useful approach for practical regression-based prediction systems.

Time-series forecasting methods such as ARIMA and SARIMA are also widely used for temporal weather prediction. More advanced deep learning architectures such as LSTM and Transformer-based models can capture temporal dependencies in sequential observations.

This project focuses on a practical machine learning implementation for temperature prediction while providing a foundation for future improvements.

5 Research Methodology

The proposed system follows a systematic machine learning methodology. The major stages are:

1. Data collection
2. Data preprocessing
3. Exploratory Data Analysis
4. Feature selection and preparation
5. Model training
6. Model evaluation
7. Application development
8. Testing and documentation

The workflow begins with weather data collection and continues through preprocessing and analysis. Relevant meteorological variables are then used as features for training the prediction model. After training, the model is evaluated using regression metrics. Finally, the trained model is integrated into a prediction application.

6 Dataset Description

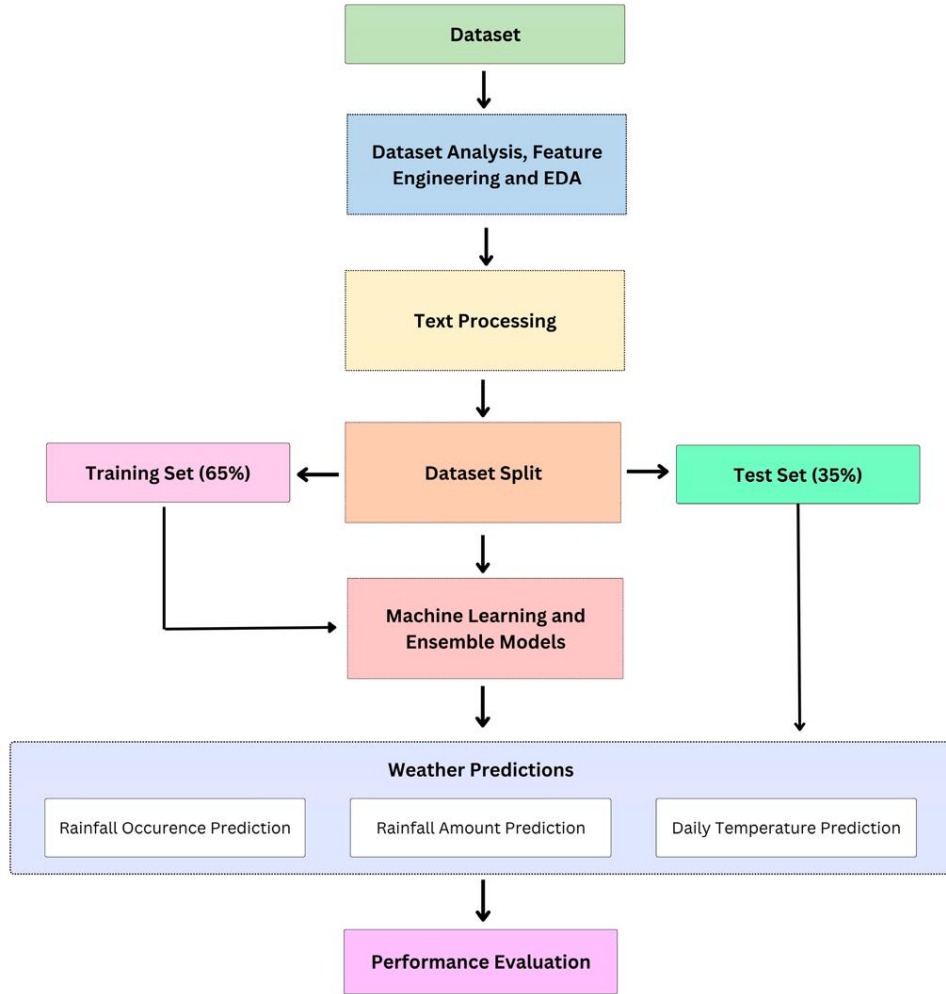


Figure 1: Weather Prediction System Workflow

The dataset used in this project contains meteorological observations required for temperature prediction. The selected input variables are humidity, wind speed, and atmospheric pressure.

The target variable is temperature.

Table 1: Variables Used for Prediction

Variable	Type	Role
Humidity	Numerical	Input Feature
Wind Speed (km/h)	Numerical	Input Feature
Pressure (millibars)	Numerical	Input Feature
Temperature (C)	Numerical	Target Variable

The project uses a weather history dataset containing the meteorological observations required by the prediction model.

7 Data Preprocessing

Data preprocessing is an important step in machine learning because raw data may contain missing values, duplicate observations, inconsistent formats, or other quality issues.

The preprocessing workflow includes:

- Loading the weather dataset.
- Inspecting the dataset structure.
- Checking for missing values.
- Selecting relevant numerical features.
- Separating input variables from the target variable.
- Splitting the data into training and testing sets.

The input feature vector can be represented as:

$$X = [H, W, P] \tag{1}$$

where H represents humidity, W represents wind speed, and P represents atmospheric pressure. The target temperature is represented by T .

The machine learning model learns a function:

$$T = f(H, W, P) \tag{2}$$

which is then used to estimate temperature for new input values.

8 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the weather dataset.

The analysis includes:

- Dataset shape and structure analysis.
- Statistical summary of numerical variables.
- Missing-value analysis.
- Feature distribution analysis.

- Correlation analysis.
- Visualization of relationships between input features and temperature.

Correlation analysis is useful for understanding the strength and direction of relationships among meteorological variables. The results of EDA can also help determine which features are useful for model development.

9 Feature Engineering

Feature engineering involves selecting and preparing variables so that machine learning algorithms can learn useful patterns effectively.

For this system, the selected meteorological features are:

- Humidity
- Wind Speed
- Atmospheric Pressure

These features are provided to the trained model as numerical inputs.

If additional weather variables become available, the system can be extended by incorporating rainfall, cloud cover, visibility, historical temperature, geographic location, and time-related features.

10 Machine Learning Model

The prediction problem is a supervised regression task because the target variable, temperature, is continuous.

A **Random Forest Regressor** is used as the machine learning model. Random Forest is an ensemble learning method that combines predictions from multiple decision trees to produce a regression output.

In this implementation, the model is initialized with:

```
RandomForestRegressor(random_state=42)
```

The dataset is divided into training and testing sets using an 80:20 ratio. The random state is set to 42 to make the data split reproducible.

The trained model learns the relationship between humidity, wind speed, and pressure and the corresponding temperature.

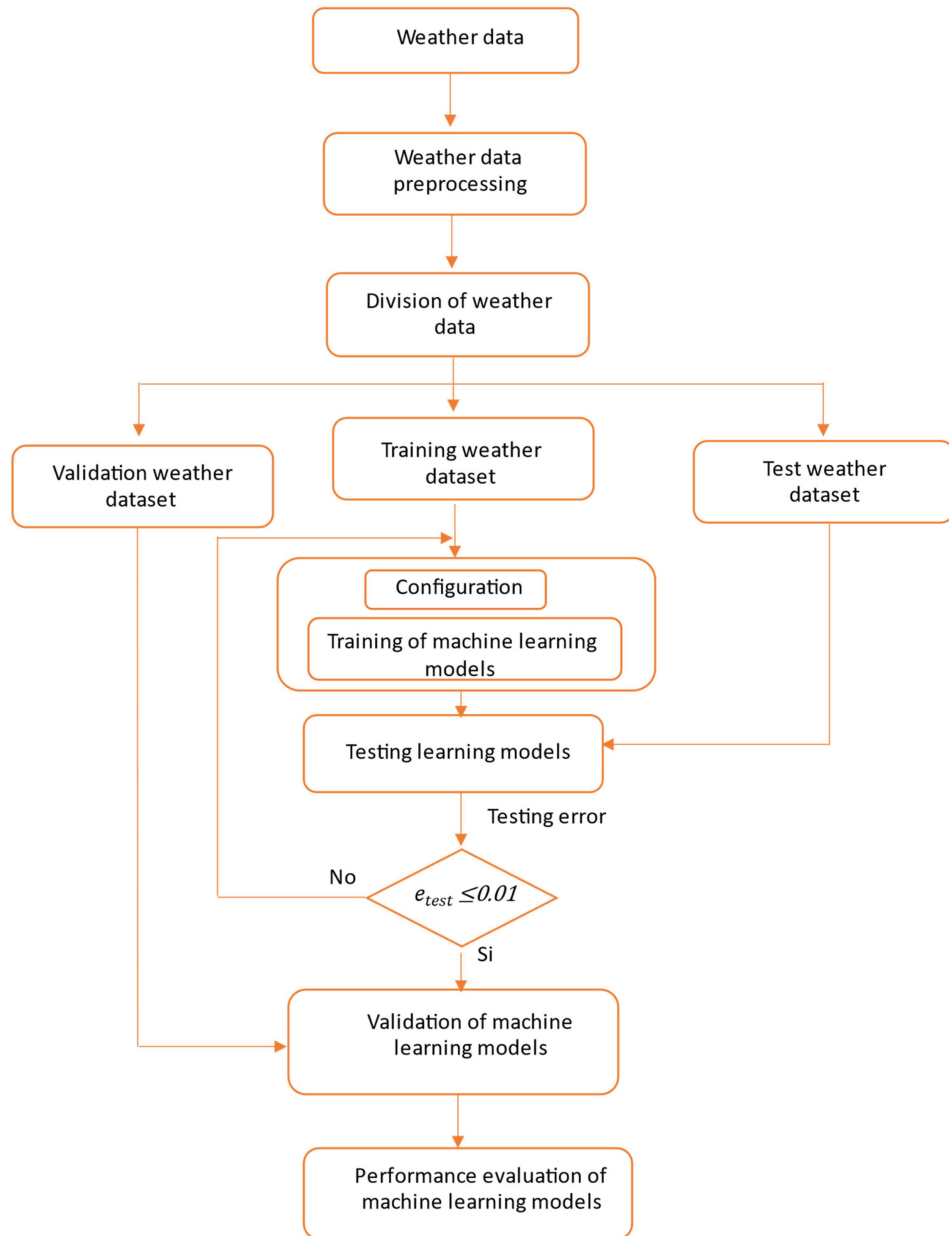


Figure 2: Random Forest Model Architecture

11 Experimental Setup

The system was developed using Python as the primary programming language. Visual Studio Code was used as the development environment.

The implementation uses Python libraries for:

- Data processing
- Numerical computation
- Machine learning
- Model evaluation
- Model serialization
- Application development

The experimental workflow consisted of:

1. Loading the weather dataset.
2. Preparing the input and target variables.
3. Dividing the data into training and testing sets.
4. Training the Random Forest Regressor.
5. Generating predictions on test data.
6. Calculating evaluation metrics.
7. Saving the trained model as `weather_model.pkl`.
8. Integrating the model into the application.

12 Evaluation Metrics

The regression model is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 .

12.1 Mean Absolute Error

MAE measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

12.2 Mean Squared Error

MSE calculates the average squared prediction error.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

12.3 Root Mean Squared Error

RMSE is the square root of MSE.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

12.4 Coefficient of Determination

The coefficient of determination, R^2 , measures how well the model explains the variation in the target variable.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (6)$$

13 Results

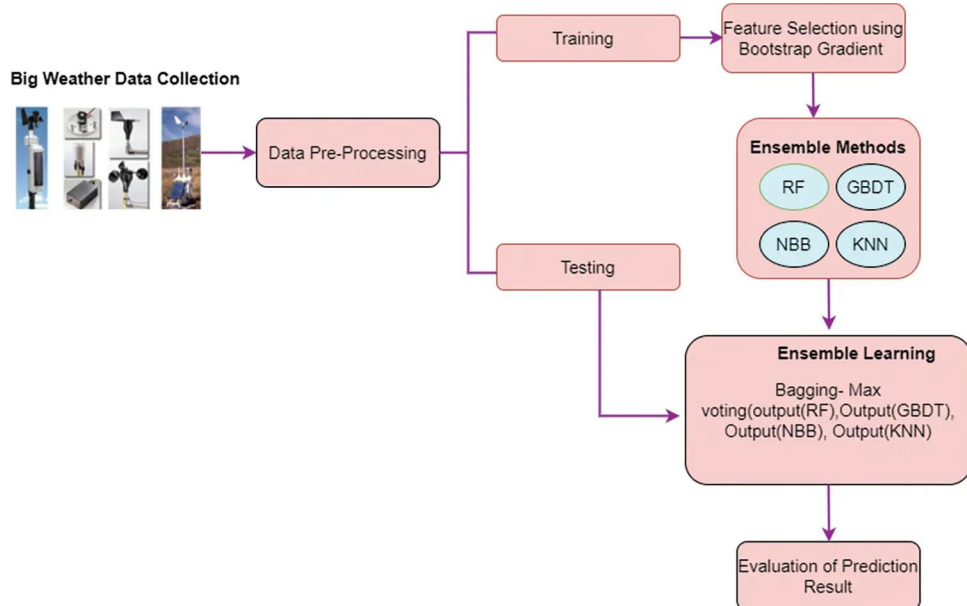


Figure 3: Random Forest Model Architecture

The trained Random Forest Regressor was evaluated using the test dataset. The following are the actual performance values obtained from the model.

Table 2: Model Performance

Metric	Value
MAE	4.8610926250
MSE	39.1626886051
RMSE	6.2580099556
R^2 Score	0.5750542902

The MAE of approximately 4.86 indicates the average absolute prediction error. The RMSE is approximately 6.26, while the R^2 score is approximately 0.575. These values provide a quantitative measure of the model’s predictive performance on the test data.

14 Application Development

The trained model was integrated into a user-facing weather prediction application.

The application allows users to enter:

1. Humidity
2. Wind Speed
3. Atmospheric Pressure

The basic application workflow is:

1. User enters humidity.
2. User enters wind speed.
3. User enters atmospheric pressure.
4. The application passes these values to the trained model.
5. The model generates the predicted temperature.
6. The predicted temperature is displayed to the user.

The trained model is stored as `weather_model.pkl`, allowing the application to load the trained model without retraining it every time.

15 Discussion

The developed system demonstrates the application of machine learning to weather-related temperature prediction. By using historical meteorological data, the Random Forest model can learn relationships among weather variables and generate predictions for new observations.

The obtained R^2 value of approximately 0.575 indicates that the model captures a meaningful portion of the variation in the target temperature. However, the prediction error metrics also show that there is room for improvement.

The effectiveness of the system depends on the quality and quantity of the training data, the selected features, the machine learning algorithm, and the evaluation methodology.

16 Limitations

The current system has several limitations:

- Prediction accuracy depends strongly on the quality and coverage of the available dataset.
- The system uses a limited set of meteorological variables.
- Weather conditions can be affected by complex spatial and temporal patterns that may not be completely captured by the current model.
- Extreme weather events may require specialized forecasting models.
- Model performance may vary when applied to locations or weather conditions that are significantly different from the training data.

17 Future Scope

The proposed system can be extended in several ways:

- Incorporating larger and more diverse meteorological datasets.
- Adding rainfall, cloud cover, visibility, and other weather variables.
- Implementing time-series forecasting models such as LSTM and Transformer-based models.
- Integrating satellite imagery and remote sensing data.

- Developing explainable AI capabilities to understand model decisions.
- Deploying the application through a scalable backend architecture.
- Extending the system into an intelligent weather research assistant using Generative AI and Retrieval-Augmented Generation.

18 Conclusion

This research presented the development of an AI-based Weather Prediction System using machine learning techniques. The proposed approach follows a complete workflow consisting of data preprocessing, exploratory data analysis, feature preparation, model training, evaluation, and application development.

The system uses humidity, wind speed, and atmospheric pressure to predict temperature using a Random Forest Regressor. The model achieved an MAE of 4.8611, an MSE of 39.1627, an RMSE of 6.2580, and an R^2 score of 0.5751 on the test dataset.

The system demonstrates how meteorological parameters can be used to develop a temperature prediction model and integrate it into an accessible application. Although the current implementation has limitations, it provides a practical foundation for future development involving advanced machine learning, deep learning, time-series forecasting, satellite data, explainable AI, and Generative AI technologies.

References

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